

Proposed Solution Report: Context-Aware Sleep Stage Classification and Anomaly Detection

1. Introduction

Sleep disorders, such as irregular sleep stages and sleep apnea, affect millions globally. Detecting anomalies in sleep patterns is critical for early diagnosis and intervention. Traditional methods for sleep stage classification often rely on large datasets, but data for rare sleep events is scarce.

This project proposes a beginner-friendly but novel approach to sleep stage classification and anomaly detection using a hybrid model that combines CNN, LSTM, and Transformer architectures. By incorporating attention mechanisms and personalized anomaly detection, this system aims to improve classification performance while requiring minimal labeled data. Furthermore, the system will provide explainable insights using attention visualization, making it easier to interpret the results.

2. Problem Statement

Sleep data is inherently complex due to long-term dependencies and variability between individuals. Detecting rare anomalies, such as transitions between REM stages or sleep apnea events, is particularly challenging because:

- Labeled data for these rare events is limited.
- Traditional models, such as LSTM or CNN, may fail to capture both local and global temporal patterns adequately.
- Differences in individual sleep patterns make generalized models less accurate.

This project addresses these challenges by developing a hybrid, context-aware model that leverages attention, multi-modal features, and personalized anomaly scoring to enhance accuracy and interpretability.

3. Objectives

The primary objectives of this project are:

1. Perform sleep stage classification using a hybrid CNN-LSTM-Transformer model.
2. Detect rare sleep anomalies, such as REM irregularities and fragmented sleep, with minimal labeled data.
3. Incorporate attention mechanisms to provide explainable insights into the model's predictions.
4. Combine raw time-series signals with handcrafted features (FFT, spectral power) to improve detection accuracy.
5. Implement personalized anomaly detection, flagging unusual sleep events based on individual subject data.

4. Proposed Solution

4.1 Data Preprocessing

- **Multi-modal Data:** Use EEG, ECG/PPG, and accelerometer signals from the Sleep-EDF and PhysioNet Sleep Apnea datasets.
- **Handling Missing Data:** Missing values will be addressed through imputation or removal, depending on the extent of the missing data.
- **Feature Normalization:** Normalize features to bring them to a comparable scale.
- **Handcrafted Features:** In addition to raw signals, extract features like FFT, spectral power, and other statistical metrics to enhance model performance.

4.2 Model Architecture

The model follows a hybrid CNN-LSTM-Transformer pipeline:

- **1D-CNN:** Extracts local temporal patterns from raw sleep signals, such as short bursts in EEG indicating REM or deep sleep.
- **LSTM Layer:** Captures short-term temporal dependencies between consecutive time steps.

- **Transformer Encoder:** Captures long-range dependencies in the sleep data, enabling context-aware anomaly detection.
- **Attention Mechanism:** Attention weights highlight which parts of the signal are most influential for classification, providing interpretability.
- **Output Layer:** A dense layer outputs sleep stage probabilities.

4.3 Novelty Features

- **Hybrid Model Architecture:** Combining CNN, LSTM, and Transformer improves accuracy compared to baseline models.
- **Feature Fusion:** Merges raw signals with handcrafted features, providing a richer input representation.
- **Personalized Anomaly Detection:** Anomalies are scored relative to each subject's baseline sleep pattern, enhancing the detection of individual-specific events.
- **Explainable AI:** Visualizing attention maps enables the identification of time segments that are crucial for sleep stage classification and anomaly detection.

5. Practical Implementation Steps

Step 1: Baseline Model

- Train a basic 1D-CNN → LSTM → Dense model using EEG signals.
- Evaluate performance using accuracy and F1-score.

Step 2: Transformer Integration

- Add a Transformer encoder layer after LSTM to capture long-range dependencies.
- Compare the model's performance with the baseline model.

- Visualize attention weights to interpret which segments of the data are most influential.

Step 3: Adding Novelty

- Introduce spectral and statistical features alongside raw signals.
- Implement personalized anomaly scoring to detect rare sleep events.
- Generate attention heatmaps to visualize which parts of the sleep session are most critical for detection.

6. Evaluation Metrics

- **Sleep Stage Classification:** Accuracy, F1-score, Confusion Matrix.
- **Anomaly Detection:** Precision, Recall, F1-score for rare events.
- **Explainability:** Use attention visualizations to identify critical time segments for sleep stage classification and anomaly detection.

7. Datasets

1. **Sleep-EDF Dataset:** Annotated sleep stages for EEG and physiological signals.
2. **PhysioNet Sleep Apnea Dataset:** A dataset for detecting sleep apnea and other rare anomalies.
3. **Optional Kaggle Sleep Disorder Datasets:** Additional datasets for anomaly detection.

8. Timeline (5 Months, Beginner-Friendly)

Month	Tasks
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- 1 Collect and preprocess data; extract handcrafted features; exploratory analysis.
- 2 Implement baseline CNN-LSTM model; train and evaluate sleep stage classification.
- 3 Add Transformer encoder; integrate attention mechanism; train hybrid model.
- 4 Implement personalized anomaly detection; combine raw + spectral features; evaluate.
- 5 Visualize attention maps; finalize model; document results and write report.

9. Expected Outcomes

1. A robust sleep stage classification system with improved accuracy via the hybrid CNN-LSTM-Transformer model.
2. Ability to detect rare sleep anomalies with minimal labeled data.
3. Explainable attention visualizations for model interpretability.
4. Personalized anomaly detection that is adaptable to individual subjects.

10. Conclusion

This project leverages the combination of CNN, LSTM, and Transformer models, along with attention-based explainability and personalized anomaly detection, to provide a practical and beginner-friendly solution for sleep stage classification and anomaly detection. By integrating multi-modal signals and handcrafted features, the system improves accuracy and interpretability. The project will be completed within five months, making it suitable for beginners, and it lays the groundwork for future advancements in sleep data analysis.